

How to Run the App

Step-by-step guide to run Chatbot Platform locally and in production.

What you need

Requirement	Notes
Python 3.11+	3.14 works; see Python 3.14 note below
Node.js 18+	Comes with npm 9+
LLM API key	From OpenAI , OpenRouter , Z.AI , Grog , or any OpenAI-compatible provider

The app runs two processes:

Service	URL	Port
Backend (FastAPI)	http://localhost:8000	8000
Frontend (Vite)	http://localhost:5173	5173

Option A — One command (recommended)

From the project root:

```
chmod +x scripts/dev.sh  # first time only
./scripts/dev.sh
```

On first run, the script:

- 1. Creates `backend/.venv` and installs Python dependencies
- 2. Starts the API on port **8000**
- 3. Starts the UI on port **5173**

Open <http://localhost:5173> in your browser.

Press `Ctrl+C` in the terminal to stop both servers.

Note: Run `npm install` in `frontend/` once before using `dev.sh` if you have not installed frontend dependencies yet.

Option B — Manual setup

Use two terminals if you prefer to run backend and frontend separately.

Terminal 1 — Backend

```
cd backend

# 1. Virtual environment
python3 -m venv .venv
source .venv/bin/activate          # Windows: .venv\Scripts\activate

# 2. Dependencies
pip install -r requirements.txt
# On Python 3.14 if pip build fails:
# PY03_USE_ABI3_FORWARD_COMPATIBILITY=1 pip install -r requirements.txt

# 3. Environment file
cp .env.example .env

# 4. Start API
uvicorn app.main:app --reload --port 8000
```

Verify the backend:

```
curl http://localhost:8000/api/health
# {"status":"ok"}
```

Swagger docs: <http://localhost:8000/docs>

Terminal 2 — Frontend

```
cd frontend
npm install
npm run dev
```

Open <http://localhost:5173>

The Vite dev server proxies `/api` to `http://localhost:8000`, so the frontend talks to the backend without extra CORS setup.

Environment variables

Copy the example file and edit if needed:

```
cp backend/.env.example backend/.env
```

Variable	Default	Purpose
SECRET_KEY	change-me-to-a-random-secret-key	JWT signing — change before production
DATABASE_URL	sqlite::///./chatbot.db	Database (SQLite file created in backend/)
CORS_ORIGINS	http://localhost:5173,http://localhost:3000	Allowed frontend origins

Local development works with the defaults. No LLM keys go in **.env** — you configure those per agent in the UI.

First-time usage

1. Go to <http://localhost:5173/register> and create an account
2. Log in at **/login**
3. On the dashboard, click **New Agent**
4. Open the agent → **LLM Settings** tab and configure your provider (see examples below)
5. Click **Save**
6. Switch to the **Chat** tab and send a message

Optional:

- **Prompts** tab — add named instruction templates
- **Files** tab — upload files (OpenAI provider only)
- Clear chat history from the Chat tab

LLM provider setup (in the UI)

Configure these in each agent’s **LLM Settings** tab after the app is running.

OpenAI

Field	Value
Provider	OpenAI
API Key	sk- ...
Model	gpt-4o-mini

OpenRouter

Field	Value
Provider	OpenRouter
Base URL	https://openrouter.ai/api/v1
API Key	your OpenRouter key
Model	openai/gpt-4o-mini

Z.AI (GLM)

Field	Value
Provider	Custom
Base URL	https://api.z.ai/api/paas/v4
API Key	your Z.AI key
Model	glm-5.1 or glm-5.2

Groq

Field	Value
Provider	Custom
Base URL	https://api.groq.com/openai/v1
API Key	your Groq key
Model	llama-3.3-70b-versatile

Local (Ollama)

Field	Value
Provider	Custom
Base URL	http://localhost:11434/v1
API Key	any placeholder (e.g. ollama)
Model	your pulled model name

Production

Backend

```
cd backend
source .venv/bin/activate
uvicorn app.main:app --host 0.0.0.0 --port 8000
```

Use PostgreSQL instead of SQLite:

```
DATABASE_URL=postgresql://user:password@host:5432/chatbot
SECRET_KEY=<long-random-secret>
CORS_ORIGINS=https://your-frontend-domain.com
```

Frontend

```
cd frontend
npm run build
```

Deploy the `frontend/dist` folder to Vercel, Netlify, or similar. Point `/api` at your backend (reverse proxy or env-based API URL).

Preview the production build locally:

```
npm run preview
```

Troubleshooting

Problem	Fix
<code>pip install</code> fails on Python 3.14	<code>PY03_USE_ABI3_FORWARD_COMPATIBILITY=1 pip install -r requirements.txt</code>
Frontend shows network errors	Confirm backend is on port 8000 and frontend on 5173
<code>npm run dev</code> fails	Run <code>npm install</code> in <code>frontend/</code> first
Chat says API key not configured	Set key in LLM Settings and click Save
401 Unauthorized	Log in again — JWT may have expired (default: 7 days)
Stale or corrupt data	Stop backend, delete <code>backend/chatbot.db</code> , restart (tables recreate on startup)
Port already in use	Stop the other process or change ports in <code>uvicorn</code> / <code>vite.config.ts</code>

Python 3.14

Some native dependencies may need:

```
export PY03_USE_ABI3_FORWARD_COMPATIBILITY=1
pip install -r requirements.txt
```

The `dev.sh` script sets this automatically on first install.

Quick reference

```
# Start everything
./scripts/dev.sh

# Backend only
cd backend && source .venv/bin/activate && uvicorn app.main:app --reload --port 8000

# Frontend only
cd frontend && npm run dev

# Health check
curl http://localhost:8000/api/health

# Reset database
rm backend/chatbot.db    # with backend stopped
```

See also

- [Main README](#) — project overview and API reference
- [Architecture & design](#) — how the system is built